VentureMiner AI

Opportunity Discovery Engine

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Document 14 — Opportunity Discovery Engine

> *How the platform finds candidate opportunities from raw signals. The discovery engine is the upstream of the entire pipeline; the quality of everything downstream depends on it.*

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1. Purpose & Scope

This document defines the discovery engine — the system that takes a seed (or watchlist) and produces a list of candidate opportunities. It is the entry point to the AI pipeline.

2. Discovery flow

```
seed + window + depth
→ query construction
→ fan-out to sources
→ signal capture
→ dedupe + cluster
→ novelty score
→ rank + filter
→ present to user
```

Detailed in Document 03 §4 and Document 08 §3.

3. Sources

Reddit	Public + OAuth	OAuth	Free
X (Twitter)	API v2	OAuth	$$
Hacker News	Public	—	Free

Reddit	Public + OAuth	OAuth	Free
GitHub trending	Public API	Optional OAuth	Free
Google Trends	Public	—	Free
G2 reviews	Public	—	Free
Product Hunt	Public	OAuth	Free
App Store / Play Store	Public RSS	—	Free
LinkedIn (post-MVP)	API	OAuth	$$$
Industry data packs	Curated	—	$$$

Each source has a connector (Document 13) and a configurable weight.

4. Query construction

The discovery planner (AGT-DISC-PLANNER) generates a list of source-specific queries from a seed. For example, given "AI for solopreneurs":

- Reddit: solopreneur indie hacker AI tools (last 30d, top 200)
- X: solopreneur AI (last 30d, top 200 by engagement)
- G2: category ai-assistants, sort by trending
- Google Trends: solopreneur AI, indie AI tools, …

5. Signal capture

Each connector returns a normalized DiscoveryHit:

```
{
"id": "...",
"title": "...",
"url": "...",
"summary": "...",
"source": "reddit",
"source_meta": { "subreddit": "...", "score": 412, "comments": 53 },
"captured_at": "2026-07-20T12:00:00Z",
"freshness": "live",
"embedding": [ ... ]
}
```

Signals are stored in discovery.discovery_hit and indexed in the RAG corpus (Document 10).

6. Clustering & dedupe

- Embedding-based clustering (Document 09 §5).
- Intra-cluster similarity ≥ 0.85 → merge.

7. Novelty scoring

Each candidate receives a novelty score based on:

- Frequency in last 90d (rising vs. flat).
- Sentiment shift (positive vs. negative trend).
- Source diversity (more sources = more cross-validated).
- Uniqueness vs. existing opportunities in the workspace (cosine distance).

Score range: 0–10.

8. Filtering & ranking

- Filter: must have ≥ 2 distinct sources (or 1 high-trust source for Quick depth).
- Filter: must have at least 1 pain point, trend, or feature request (not pure news).
- Rank: composite of novelty + source diversity + freshness.

9. Continuous discovery

- Always-on mode (Team+, post-MVP) runs the planner on a schedule per workspace.
- A watchlist defines seeds; planner produces daily runs; user is notified of high-novelty hits.
- Quiet hours can be configured.

10. Cost & scale

- v1: 5k discovery runs/day, 200k hits/day, 50k candidate opportunities/day.
- v2: 50k discovery runs/day.

11. Quality evaluation

- Hit precision: of hits surfaced, what fraction leads to a pinned opportunity?
- Cluster purity: of clusters, what fraction are clean (single concept)?
- Recall vs. baseline: is the engine finding opportunities that a human researcher finds?
- Eval set: 100 known opportunities + 100 known non-opportunities.

12. Failure modes

Source API down	Skip; mark partial; user notified.
Source rate-limited	Backoff; cap; rotate keys.
Embedding failure	Use lexical clustering fallback.
Low hit yield	Surface notice; suggest query relaxation.
High hit yield	Cap and rank; offer to broaden depth.

13.1 Revision history

v0.5	2026-07-20	Doc Team	All sections drafted
v1.0	2026-07-20	Doc Team	First approved version

13.2 Cross-references

- Application Flow: Document 03 §4.
- Multi-Agent: Document 08.
- Agent Specs: Document 09.
- RAG: Document 10.
- Plugin Architecture: Document 13.

End of Document 14 — Opportunity Discovery Engine. The quality of everything downstream depends on this engine's discipline.